

# Attosecond-Resolution Er:Fiber-Based Optical Frequency Comb \*

YAN Lu-Lu(闫露露)<sup>1</sup>, ZHANG Yan-Yan(张颜艳)<sup>1</sup>, ZHANG Long(张龙)<sup>1</sup>,  
FAN Song-Tao(樊松涛)<sup>1,2</sup>, ZHANG Xiao-Fei(张晓斐)<sup>1</sup>, GUO Wen-Ge(郭文阁)<sup>3</sup>,  
ZHANG Shou-Gang(张首刚)<sup>1</sup>, JIANG Hai-Feng(姜海峰)<sup>1\*\*</sup>

<sup>1</sup>Key Laboratory of Time and Frequency Primary Standards, National Time Service Center,  
Chinese Academy of Sciences, Xi'an 710600

<sup>2</sup>Graduate University of Chinese Academy of Sciences, Beijing 100039

<sup>3</sup>School of Science, Xi'an Shiyou University, Xi'an 710065

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*Highly stable frequency-controlled optical frequency combs are key elements of many applications in time-frequency and optical-metrology domains. In this work, we demonstrate a highly stable frequency-controlled erbium-fiber-based optical frequency comb system. Its repetition rate is phase-stabilized to a continuous-wave laser with both an intra-cavity electro-optic modulator and a piezo-transducer; while the carrier-envelope-offset frequency is phase-locked to a radio-frequency signal generator by controlling the pump power. In-loop relative frequency stabilities of the comb are below  $1 \times 10^{-16}$  at 1 s, and integrate down to low  $10^{-20}$  level at  $10^4$  s. The corresponding timing uncertainties are 100–200 as over the full measurement range.*

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Optical frequency combs play important roles in many research fields such as time and frequency metrology,<sup>[1]</sup> attosecond science,<sup>[2]</sup> high resolution spectroscopy,<sup>[3]</sup> ultra-stable microwave generation,<sup>[4]</sup> high precision ranging,<sup>[5]</sup> time and frequency transfer,<sup>[6]</sup> and in fundamental physics research.<sup>[7]</sup> There are kinds of combs in terms of gain media. The most popular combs are Ti:sapphire-based,<sup>[8]</sup> Er:fiber-based<sup>[9]</sup> and Yb:fiber-based<sup>[10]</sup> ones. Among these combs, Er:fiber-based optical combs are of great advantage to system compactness and reliability. In particular, it directly covers telecommunication wavelength, which is convenient to many applications.

Since the optical frequency combs were invented in the late 1900s,<sup>[11,12]</sup> they have resulted in the development of kinds of optical atomic clocks with frequency instability well below the best microwave atomic clocks.<sup>[13,14]</sup> Recently, frequency stability of optical clocks reached  $\sim 3 \times 10^{-16}$  at 1 s and low  $10^{-18}$  level at  $10^4$  s,<sup>[15,16]</sup> and frequency stability of the future nuclear atomic clock will be even better.<sup>[17]</sup> To precisely measure and apply these ultra-stable frequency references, optical combs should have lower additional noise effect than the clocks. Usually, the additional noise level is evaluated with the frequency stability of the comb's repetition rate ( $f_r$ ) and carrier-envelope-offset frequency ( $f_{\text{ceo}}$ ). Note that  $f_r$  and  $f_{\text{ceo}}$  determine frequencies of all optical modes with relation of  $\nu_n = f_{\text{ceo}} + n \times f_r$ , where  $n$  is an integer, and  $\nu_n$  is frequency of the  $n$ th optical mode. Now, the residual frequency stability of optical frequency combs at the state-of-the-art level is below  $1 \times 10^{-17}$  at 1 s.<sup>[18]</sup>

The National Time Service Center (NTSC) has developed Er:fiber-based optical frequency combs for frequency metrology since 2013. Last year, we developed an Er:fiber comb with an intra-cavity electro-optic modulator (EOM), enabling fast repetition rate control.<sup>[19]</sup> The comb has a repetition rate of about 209 MHz; the comb's frequency stability is in low  $10^{-13}$  level at 1 s and low  $10^{-15}$  level at  $10^4$  s.<sup>[20]</sup> In this Letter, we present our latest Er:fiber-based comb. Compared with the previous comb, both electronic and optic systems of this one are improved. By taking advantage of high resolution measurement in the optical domain, in-loop frequency stabilities of this comb are 2–3 orders of magnitude below that of the previous one. Furthermore, the comb's repetition rate is higher, and the direct output power is higher. This is helpful for obtaining high signal-to-noise ratio (SNR) optical beatnote in frequency metrology applications.

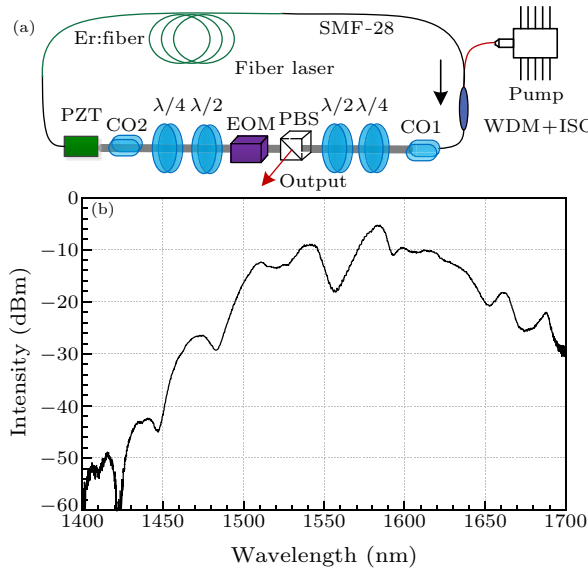
Setup of the Er:fiber mode-locked laser is shown in Fig. 1(a). The laser's cavity contains a 34.5 cm SMF-28 fiber, a 43.5 cm of 110 dB/m erbium-doped gain fiber, a free space part including two collimators, four wave plates, a polarization beam splitter and a lithium niobate crystal (8 mm long, 3.2 mm thick) acting as a fast cavity length controller. At 1550 nm wavelength, we estimate the net dispersion of the cavity is about  $-2000 \text{ fs}^2$ , the net dispersion of the fiber is about  $-2310 \text{ fs}^2$ , the dispersion of EOM is about  $800 \text{ fs}^2$ , and the net dispersion of other components is about  $-490 \text{ fs}^2$ . The pump sources are two single-mode 980 nm pigtailed diode lasers. Figure 1(b) shows the optical spectrum of the mode-locked laser in loga-

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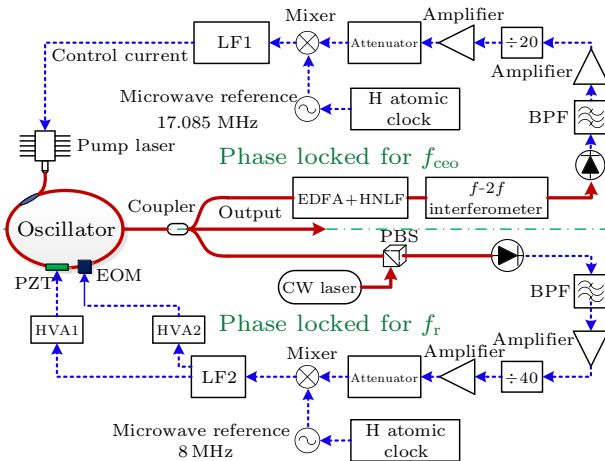
\*\*Corresponding author. Email: haifeng.jiang@ntsc.ac.cn

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rithmic scale. The 10 dB bandwidth is approximately 120 nm centered at  $\sim 1570$  nm. The repetition rate is about 232 MHz. The output optical power is about 180 mW under 1 W pumping power. The laser output is separated into three parts. Two parts are used to detect  $f_{\text{ceo}}$  and  $f_r$ , while the other is used for applications.



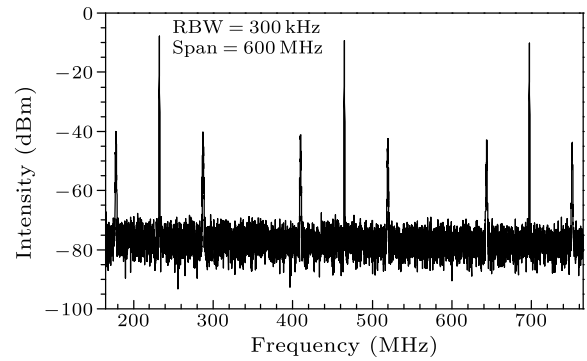
**Fig. 1.** (a) Schematic diagram of our home-made Er:fiber-based ring laser, and (b) optical spectrum of the mode-locked laser in logarithmic scale. The thick solid lines represent free-space paths, and the thin solid lines and red solid lines represent fiber paths. Here  $\lambda/2$ : half-wave plate;  $\lambda/4$ : quarter-wave plate; WDM: wavelength division multiplexer; ISO: isolator; PBS: polarization beam splitter; CO: collimator; PD: photodetector; PZT: piezo-electric transducer; and EOM: electro-optic modulator.



**Fig. 2.** Frequency control system for  $f_{\text{ceo}}$  and  $f_r$  between the comb and a continuous wave reference laser. The red solid lines represent optical paths, and the blue dotted lines represent electrical paths. LF: loop filter; HVA: high voltage amplifier; BPF: band-pass filter;  $\div 20$ : 20-times frequency divider;  $\div 40$ : 40-times frequency divider; EDFA: erbium-doped fiber amplifier; and HNLF: highly nonlinear fiber.

As shown in the upper part of Fig. 2,  $f_{\text{ceo}}$  is obtained by using the  $f-2f$  interference method. We

pre-chirp, amplify and width-compress  $\sim 50$  mW laser pulse train to be 300 mW with  $\sim 60$  fs width. Then, a spliced highly nonlinear fiber (47 cm with a nonlinear coefficient of  $\sim 24$  W/km) produces an octave super-continuum (SC) spanning from 1  $\mu\text{m}$  to 2  $\mu\text{m}$ . Note that the nonlinear EDFA contains a pre-chirp fiber (103 cm SMF-28), a gain fiber (84 cm Er110) and a de-chirp fiber (34 cm Hi-1060 and 53 cm SMF-28). After that, the SC light is fed into a common-path  $f-2f$  interferometer. The value of  $f_{\text{ceo}}$  was detected by an InGaAs photodetector with an SNR of about 45 dB under 300 kHz resolution. We pick up  $f_{\text{ceo}}$  signal at  $\sim 341.7$  MHz with a band-pass filter, and amplify it to be 1–5 dBm. To enhance the system robustness, we divide frequency  $f_{\text{ceo}}$  by a factor of 20 before producing a phase-error signal by comparing its phase to the phase of a frequency reference. The 17.085 MHz reference signal is generated with a synthesizer synchronized to a hydrogen atomic clock. Here  $f_{\text{ceo}}$  is stabilized onto the reference frequency by feeding-back the error signal to a pump diode's current control port via a proportional integrator. The coefficient of  $f_{\text{ceo}}$  versus pump current is about 0.2 MHz/mA.

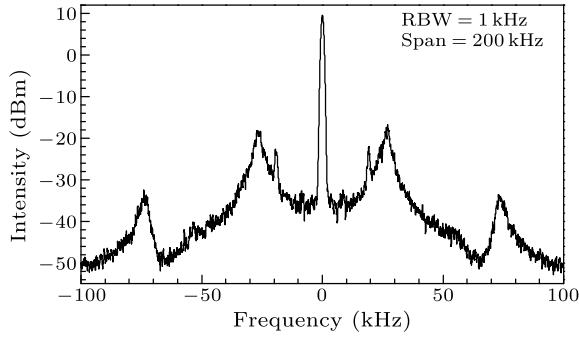


**Fig. 3.** The frequency spectrum of the beatnote between the comb and a cw laser.

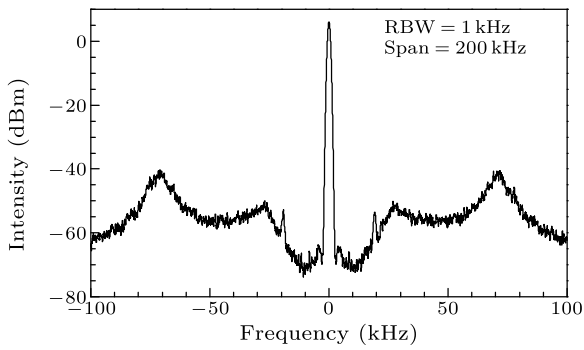
Once  $f_{\text{ceo}}$  is fixed, frequencies of the comb are determined by only one parameter, i.e.,  $f_r$ . Thus we can stabilize the comb's frequencies onto a cw laser by phase-locking one mode of the combs on the cw laser.

The lower part of Fig. 2 shows the scheme of stabilizing  $f_r$  onto a cw laser. The beat note between the comb and the cw laser (1–2 mW, 1550.5 nm) was detected with a free-space photodiode with an SNR of  $\sim 35$  dB under 300 kHz resolution as shown in Fig. 3. We select the beating frequency at about 320 MHz and amplify it to 1–5 dBm. Then, we employed a 40-time frequency divider to divide the selected signal so as to improve robustness. Finally, frequency divided beat-note signal at 8 MHz is stabilized onto a reference frequency by controlling the PZT and the EOM, yielding a long-term tight phase-locking. Note that the EOM has an input range from  $-200$  V to  $+200$  V resulting in an  $f_r$  frequency tuning of 30 Hz; while the PZT has an

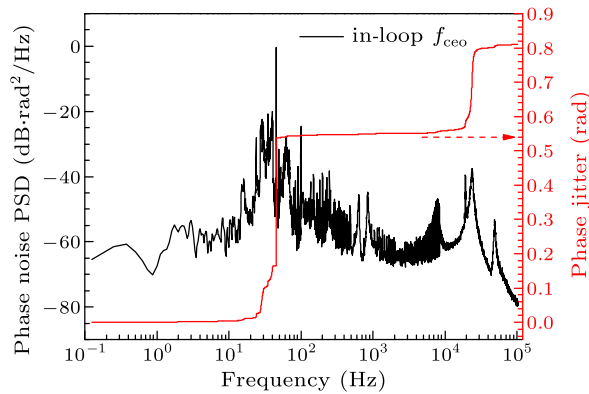
input range from 0 V to 200 V corresponding to 3 kHz frequency control.



**Fig. 4.** Frequency spectrum of frequency-divided in-loop  $f_{\text{ceo}}$ .



**Fig. 5.** Frequency spectrum of frequency-divided in-loop  $f_r$ .

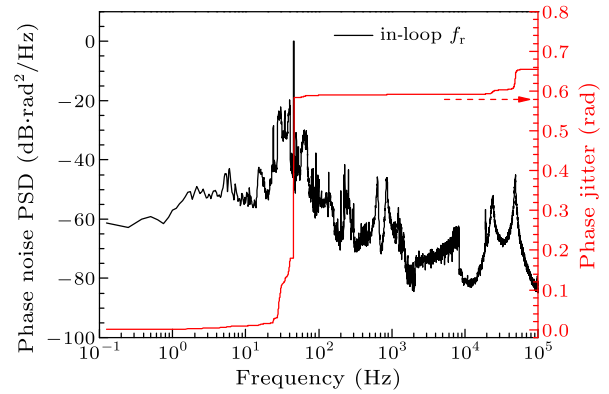


**Fig. 6.** Phase noise and phase jitter of in-loop  $f_{\text{ceo}}$ . PSD: power spectral density.

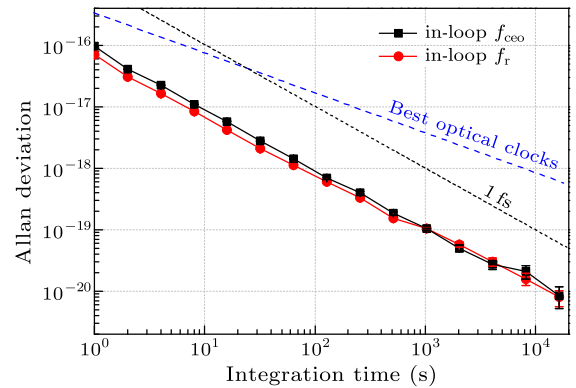
Cross-talk happens when  $f_r$  and  $f_{\text{ceo}}$  are stabilized at the same time. This leads to two control loops oscillating at the same frequency as shown in Figs. 4 and 5. To obtain a high resolution comb, we make a trade-off between two phase lock loops (PLL) by optimizing parameters of two loop filters. The control bandwidth of two loops are lower than that of the loop working independently. The servo bandwidths of  $f_{\text{ceo}}$  and  $f_r$  are estimated to be about 25 kHz and 70 kHz, respectively, indicated by gain bumps.

To evaluate the PLL performance more precisely, we measured the in-loop phase noise of  $f_{\text{ceo}}$  and

$f_r$  with a dynamic signal analyzer (SR785). The phase noise of  $f_{\text{ceo}}$  is generally in the range of  $-50 \text{ dB}\cdot\text{rad}^2/\text{Hz}$ – $-60 \text{ dB}\cdot\text{rad}^2/\text{Hz}$  except some noise bumps at 50 Hz and 25 kHz as shown in Fig. 6; while the phase noise of  $f_r$  is generally below  $-50 \text{ dB}\cdot\text{rad}^2/\text{Hz}$  except noise bumps at 50 Hz and 70 kHz as shown in Fig. 7. Phase jitters (i.e., square root of accumulated phase noise) from 1 Hz to 100 kHz are 0.8 rad and 0.65 rad for  $f_{\text{ceo}}$  and  $f_r$ , respectively. It is clear that the main contribution is from the 50 Hz noise bumps, which is supposed to be the oscillation due to PZT resonance and the cross-talk effect evoked by environmental vibrations. Therefore, significant noise rejection is possible by optimizing parameters of PZT and cross-talk.



**Fig. 7.** Phase noise and phase jitter of in-loop  $f_r$ .



**Fig. 8.** In-loop relative frequency stability of  $f_{\text{ceo}}$  (solid black square) and  $f_r$  (solid red circle).

Figure 8 shows the in-loop relative frequency stabilities of  $f_{\text{ceo}}$  and  $f_r$  (normalized with the optical frequency of  $\sim 195 \text{ THz}$ ), which are about  $1 \times 10^{-16}$  and roll down with a slope of  $1/\tau$ , and well below that of the best optical clocks.<sup>[15,16]</sup> The corresponding timing uncertainties are 100–200 as over the full measurement range. In this measurement,  $f_{\text{ceo}}$  and  $f_r$  are recorded with a dead-time frequency  $\Pi$ -type counter under the 1000-time averaging mode. Averaging reduces the high frequency noise effect like the  $\Lambda$ -type counter does.<sup>[21]</sup> Considering the noise rejection effect of about one order, the measured frequency stabilities

are consistent with the previously measured phase jitters.

In summary, we have demonstrated a home-made optical frequency comb with an intra-cavity electro-optic modulator. This comb has a direct output of 180 mW and a repetition rate of about 232 MHz. We stabilize both  $f_{\text{ceo}}$  and  $f_r$  onto an rf reference frequency and a cw laser simultaneously. The in-loop relative frequency stabilities of  $f_{\text{ceo}}$  and  $f_r$  are about  $1 \times 10^{-17}$  at 1 s and roll down to low  $10^{-20}$  level at  $10^4$  s with a slope of  $1/\tau$ . It is believed that this highly stable optical frequency comb matches the requirement of the Sr optical clock frequency measurement and application at NTSC.

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